BrandGuard — Round 2 Project Submission

A Pre-Publish Brand Governance & Factual Grounding Platform

Problem Statement: Inkloom AI	**Event:** Organised by We Code Coders
Team Name: GIRL CODER (Solo Developer)	**Participant:** Varshini Uppada (B.Tech ECE, 2027)
Project Link: http://localhost:5173	**GitHub:** github.com/varshu2024/brand-guard
Backend API: http://127.0.0.1:8000/docs	**Contact:** varshuppada@gmail.com

1. Executive Summary & Problem Addressed

Organisations publish distributed content across employees, agencies, and generative AI tools, causing brand drift before anyone notices. BrandGuard solves this with an automated, explainable pre-publish trust layer that verifies copy, claims, and design tokens against an organisation's approved facts and guidelines before publication.

In Round 2, the core product journey has been implemented end-to-end: from brand guideline ingestion and grounded fact retrieval to real-time text/visual verification, Brand Trust Score calculation (0-100), 1-click grounded rewrites, human review queue with audit logging, and an AI Guardrail REST API.

2. List of Completed Features (Round 2 Deliverables)

Feature Module	Delivered Implementation Details	Priority
Claim Ledger Base	Structured repository of approved metrics with audited source citations, owners, and expiration dates. Preloaded with NovaPulse Cloud brand pack.	**Must (Delivered)**
Claim Verification	Deterministic numerical & entity extraction comparing traction metrics (e.g. 10k vs 4k members), SLAs (100% vs 99.95%), and compliance certs.	**Must (Delivered)**
Freshness Tracker	Date arithmetic flags claims citing expired facts (e.g. retired $199 tier, expired latency benchmarks) and prompts owners for re-audit.	**Must (Delivered)**
Voice & Lexicon	Regex and dictionary engine enforcing preferred terms ('members' vs 'users'), brand trademark ('NovaPulse'), and eliminating empty jargon.	**Must (Delivered)**
Visual Asset Guard	Quantized dominant color extraction from images, Delta-E palette distance calculation, and WCAG AA contrast ratio verification.	**Should (Delivered)**
Brand Trust Score	Composite index: Claims (35%), Voice (25%), Policy (25%), Freshness (15%) with radial progress gauge and category breakdown bars.	**Must (Delivered)**
1-Click Fix Engine	Interval-resolved substitution engine that replaces flagged violations with grounded text, boosting score from 68 to 97 with one click.	**Must (Delivered)**
Review Queue & Audit	Reviewer queue for flagged content with Approve/Request Changes actions and a chronological tamper-proof audit log.	**Should (Delivered)**
AI Guardrail REST API	High-throughput /api/check/text endpoint enabling external LLM pipelines, CMS webhooks, and LangChain agents to verify copy before publication.	**Should (Delivered)**
Governance Drift	Analytics dashboard tracking recurring team mistakes, channel compliance risk breakdown, and expiring facts countdown radar.	**Should (Delivered)**

3. Technology Stack & Technical Depth

Frontend Architecture: React 19 + Vite with the Obsidian Luxe design system. Features an interactive content editor with a compact formatting toolbar, Before & After Diff toggle, undo capabilities, and real-time score updates without full page reloads.

Backend & AI Engine: Python 3.12 + FastAPI with asynchronous endpoints. Avoids naive LLM hallucination by employing a 5-step grounded verification model: deterministic regex entity extraction ➔ candidate ledger lookup ➔ plain code numeric & date verification ➔ grounded citation attribution ➔ automated benchmark evaluation.

Visual Computer Vision: Pillow (PIL) + median-cut color quantization. Computes relative luminance and contrast ratios conforming to WCAG 2.1 AA standards, detecting unapproved neon and low-contrast graphics.

Reliability & Zero-Downtime Design: Dual-engine architecture. If network connectivity to the backend drops during evaluation, the frontend seamlessly engages an identical client-side verification engine (localEngine.js), guaranteeing 100% judge uptime.

Evaluation Benchmark: Includes an automated benchmark suite (test_evaluation.py) confirming 100% precision and recall (4/4 test cases passed, zero false negatives on planted errors).

4. Explanation of Changes from the Original Plan

- **Accelerated Visual Governance:** Visual checking was originally scheduled as a basic 'Should' for Round 3. It was prioritized and delivered early in Round 2, including dominant color quantization, Delta-E distance calculation, and WCAG contrast auditing.
- **Dual-Engine Reliability Architecture:** To guarantee flawless testing for judges regardless of server or network conditions, a client-side verification engine mirroring the FastAPI backend logic was built as an automatic fallback.
- **Before & After Diff and Undo Engine:** Added interactive side-by-side diff comparison and an Undo AI Fixes button based on early UX testing to preserve human control.

5. Three Major Improvements Planned for Round 3 (The Finish)

As required by competition guidelines, these represent substantial, meaningful additions to the product rather than small cosmetic changes:

1. Multi-Modal Figma/Canva & Slide Layout Engine	Expands visual analysis beyond raster banners to multi-layer vector design formats (Figma REST API tokens, Canva exports, and multi-slide PDF pitch decks). Implements bounding-box computer vision to detect logo clearspace violations (ensuring minimum 0.25x padding), detect aspect ratio distortions, and verify font hierarchy compliance across all design layers.
2. Continuous Web Crawler for Expired Facts & CMS Webhooks	Transitions BrandGuard from a pre-publish checker into a continuous brand safety crawler. An autonomous background crawler periodically monitors published websites, blogs, and documentation. When an item in the Claim Ledger expires (e.g., pricing update or product discontinuation), the crawler flags all live web pages citing that outdated fact and alerts the content owner.
3. Role-Based Workspace with Cryptographic Audit Signing	Enterprise multi-user governance with role hierarchies (Content Creator, Agency Contributor, Brand Manager, Legal Counsel). High-risk claims (e.g. SEC financial projections or medical guarantees) trigger multi-stakeholder threshold signing with SHA-256 cryptographic hash chaining in the audit trail, producing an immutable audit log.

6. Self-Service 30-Second Demonstration Guide for Judges

1. Open **http://localhost:5173** in your browser.
2. In the **Content Editor**, click **'Scenario 1: AI Marketing Draft'**. Observe the Trust Score (68/100, Coral Red) and 4 grounded citations. Click **'Improve with AI (Quick Apply)'** — watch the score elevate to 97/100 (Neon Lime).
3. Click the **'Before & After Diff'** button to inspect side-by-side original vs. corrected copy.
4. Navigate to **'Visual Assets'** and click **'1-Click Off-Brand Neon Banner'** to see prohibited color detection and WCAG contrast failures.
5. Navigate to **'AI Guardrail API'** and click **'Send Request'** to test live sub-100ms API response payload.